Jichuan YU

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EDUCATION

Tsinghua University — Beijing, China
Ph.D. in Mechanical Engineering (Advisor: Prof. Chuxiong Hu) — Sep. 2021 – Present

Carnegie Mellon University — Pittsburgh, United States
Visiting Scholar at the Robotics Institute (hosted by Prof. Changliu Liu) — Jul. 2025 – Dec. 2025

Tsinghua University — Beijing, China
Bachelor in Mechanical Engineering (Elite Program), GPA: 3.82/4.0 (Rank: 5/133) — Sep. 2017 – Jun. 2021

McGill University — Montreal, Canada
Exchange Student in Mechanical Engineering, GPA: 4.0/4.0 — Aug. 2019 – Dec. 2019

RESEARCH INTERESTS

I have a profound interest in planning and control theories and their applications in intelligent mechatronic systems. My current Ph.D. research focuses on optimization-based motion planning and control for dual-arm robotic systems, with the goal of enabling efficient collaborative manipulation under multiple constraints while ensuring real-time responsiveness in dynamic environments. I also have prior research experiences in advanced manufacturing technologies, such as CNC machining and 3D printing.

SELECTED PUBLICATIONS († equal contribution)

[1] **J. Yu**, Z. Jin, C. Hu, et al., "Hierarchical real-time motion planning for safe multi-robot manipulation in dynamic environments," in *IEEE International Conference on Advanced Robotics and Mechatronics (ICARM)*, Tokyo, Japan, 2024. ***(Best Paper Award in Advanced Robotics)***

[2] **J. Yu†**, C. Hu†, Z. Wang, et al., "Printing three-dimensional refractory metal patterns in ambient air: Toward high temperature sensors," *Advanced Science*, vol. 10, no. 31, p. 2302479, 2023. ***(Inside Back Cover)***

[3] C. Hu, **J. Yu**, Z. Wang and Y. Zhu, "An iterative contouring error compensation scheme for five-axis precision motion systems," *Mechanical Systems and Signal Processing*, vol. 178, p. 109226, 2022.

[4] S. Lin, C. Hu, **J. Yu**, and Y. Liang, "Batch Iterative Dual Optimization for Collision-Free Robot Motion Generation," *IEEE Transactions on Industrial Informatics*, pp. 1–9, 2025.

HONORS AND AWARDS

- Second Prize for Outstanding Oral Presentation, Tsinghua University Doctoral Academic Forum — 2025
- National Scholarship for Postgraduates ***(Top 2%)*** — 2023
- Outstanding Teaching Assistant Award of Tsinghua University ***(10 recipients per year)*** — 2023
- National 1st Prize in China Mechanical Engineering Innovation Competition, Micro-Nano Sensing Technology and Intelligent Applications Category, Postgraduate Group ***(Rank: 1/88)*** — 2023
- Outstanding Graduate of Beijing — 2021

SKILLS

Programming: C/C++, Python, MATLAB
Softwares/Tools: ROS, Gazebo, SolidWorks, AutoCAD, Ansys
Domain Knowledge: Robotics, Convex Optimization, Optimal Control, Machine Learning

Last updated: Sep. 2025